

Jérémie KALFON

Computational Biologist · Researcher · Engineer · Entrepreneur

10b rue Montgallet, 75012 Paris, France

jkobject@gmail.com | +33 6 52 66 34 43 | [linkedin/jkobject](https://www.linkedin.com/in/jkobject) | [github/jkobject](https://github.com/jkobject) | [jkobject.com](https://www.jkobject.com)

PROFILE

I am mission-driven and dedicated to creating therapeutic abundance using Artificial Intelligence & High throughput biology. I have experience across research institutes and startup companies in France and the US.

EXPERIENCE

INDEPENDENT ADVISOR | SCIENTIFIC STRATEGY & EXPERTISE

Paris, FR | 2023–2026

Advised multiple companies on scientific strategy. Nucleate Leader. Member of various AI, Bio, Dev & VC communities.

WHITELABGX | TEAM LEAD, COMPUTATIONAL BIOLOGY

Paris, FR | 2022–2023

- Cellular Atlas & Knowledge Graph for target discovery & cell surface marker discovery.
- AI for cellular therapies and hit discovery.
ML, Big Data, Genomics, Structural Biology.

BROAD INSTITUTE OF MIT AND HARVARD | SR. ASSOCIATE COMP. BIOLOGIST Cambridge, MA | 2019–2022

- **DepMap Omics** — Predicting cancer dependencies from genetic sequences of cancer cell lines.
Pipeline, Big Data, Multiomics.
- **Celligner** — Alignment of cancer models to patient tumors using conditional VAEs on multi-omics data.
DL, Transcriptomics.
- **AML project** — Deciphering AML from deep epigenetic profiling with Dana Farber Cancer Institute & Boston Children's Hospital.
ML, Cancer Biology, Epigenomics.

PIPLE | CO-FOUNDER & CEO

Paris, FR | 2018–2022

Startup focused on bringing the next-generation communication platform. INSEEC incubator.
Entrepreneurship, Product Design, Full-Stack Software Development.

FLATIRON INSTITUTE, SIMONS FOUNDATION | RESEARCH ASSOCIATE

New York, NY | 2017

Member of the CalmAn project on neural spike inference from calcium imaging, under the supervision of Dmitri Chklovskii.
Research, Imaging, ML, Vision.

HIGHLIFE MED | R&D ENGINEER

Paris, FR | 2016

Biomedical startup developing the world's first no-surgery mitral heart valve replacement.
Simulation, Data Analytics.

EDUCATION

Pasteur Institute / ENS ULM

Paris, FR | 2023–2026

PH.D. ARTIFICIAL INTELLIGENCE

Harvard University

Cambridge, USA | 2019–2020

GRADUATE CERTIFICATE IN BIOLOGY

University of Kent

Canterbury, UK | 2017–2018

M.Sc. MACHINE LEARNING

ECE Paris

Paris, FR | 2013–2018

DIPLÔME D'INGÉNIEUR (M.Sc.), BIOMEDICAL ENGINEERING

Tec de Monterrey

Querétaro, MX | 2015

EXCHANGE PROGRAM

SKILLS

BigData: GCP, AWS, batch, SLURM, snakemake, terra, nextflow, HPC

Biology: Systems biology, Cancer Biology, epigenomics, Neuroscience

Datatypes: single cell, RNAseq, ST, DNAseq, ChIP-seq, ATACseq, slam-seq, BS-seq, Perturb-seq, bi-photon imaging

Code & frameworks: pytorch, python, javascript, java, R, matlab, firebase, SQL, Neo4j, nodeJS, react

Datasets: CZI, HEST, CCLE, TCGA, GTEx, Gnomad, GWASAtlas, ENCODE, Human Cell Atlas

ML: Transformers, VAEs, GNNs, CNNs, NMF, OT, optimization

Languages: English, French, Spanish

Leadership: team lead, project lead, project management, mentorship, leadership studies

PUBLICATIONS

FIRST AUTHOR

1. **scPRINT-2: Towards the next-generation of cell foundation models and benchmarks.** J. Kalfon, G. Peyré, L. Cantini. *Submitted to Nature Methods*.
2. **Towards foundation models that learn across biological scales.** J. Kalfon, L. Cantini, G. Peyré. *ICML 2025 – FM4LS, BioRxiv*.
3. **scPRINT: pre-training on 50 million cells allows robust gene network predictions.** J. Kalfon, J. Samaran, G. Peyré, L. Cantini. *Nature Communications*, 2024.
4. **Leukemia core transcriptional circuitry is a sparsely interconnected hierarchy stabilized by incoherent feed-forward loops.** J. Kalfon, T. Harada, Y. Heshmati. *BioRxiv*, 2023.
5. **A distinct core regulatory module enforces oncogene expression in KMT2A-rearranged leukemia.** J. Kalfon, T. Harada, Y. Heshmati. *Genes & Development*, 2021.

LAST AUTHOR

6. **Training foundation models on large collections of scRNA-seq data.** R. Sergei et al. (incl. J. Kalfon).

HIGHLY CITED

7. **CalmAn: an open source tool for scalable calcium imaging data analysis.** A. Giovannucci, J. Friedrich, P. Gunn, J. Kalfon et al. *ELife*, 2019.
8. **Repeat expansions confer WRN dependence in microsatellite-unstable cancers.** N. Wietmarschen et al. *Nature*, 2020.
9. **Synthetic Lethal Interaction between the ESCRT Paralog Enzymes VPS4A and VPS4B.** J.E. Neggers et al. *Cancer Research*, 2020.

5 first-author papers · 17 papers total · 1,500+ citations · 6 posters · 7 international conferences · 25 invited talks [Google Scholar](#)

AWARDS

- **AI in Bio Hackathon, Won #1** — Entrepreneur First
- **Best Poster Award** — Pasteur Institute AI in Bio Seminar
- **Spotlight Awards, 2×** — For outstanding contribution at the Broad Institute (2020, 2022)
- **ECE Innovation Awards, 2nd Place** — PiPle project (2017)
- **Berger-Levrault Innovation Awards, 2nd Place** — PiPle project (2017)

INTERESTS

Marathon Running · Triathlon · Tennis · Swimming · Cooking · Mixology · Diving & Sailing · Travel · Wood Working · Science Communication & Science Fiction